POLICY GRADIENT · SOCCER GAME

Policy gradient on the soccer game

REINFORCE, A2C and PPO, trained the standard way and by fictitious play, scored by one number: how often the trained player wins when the opponent is the best response to it.

Summary

- Against the best response, the exact solver wins 67% of games on the random board; the six learners win 5% to 14%. On the deterministic board the exact solver ties itself, so it wins 0%.
- Against a random player the learners win 61% to 96% of games.
- Starting from the 94 states where the exact answer has to mix, they win 11% to 15% against the exact Nash policy; the exact solver wins 50%.
- Fictitious-play training wins more often against a random player; neither way of training reaches the exact solver's rate against the best response.

3 seeds per learner, 1,000 games per opponent.

1. How to read the numbers

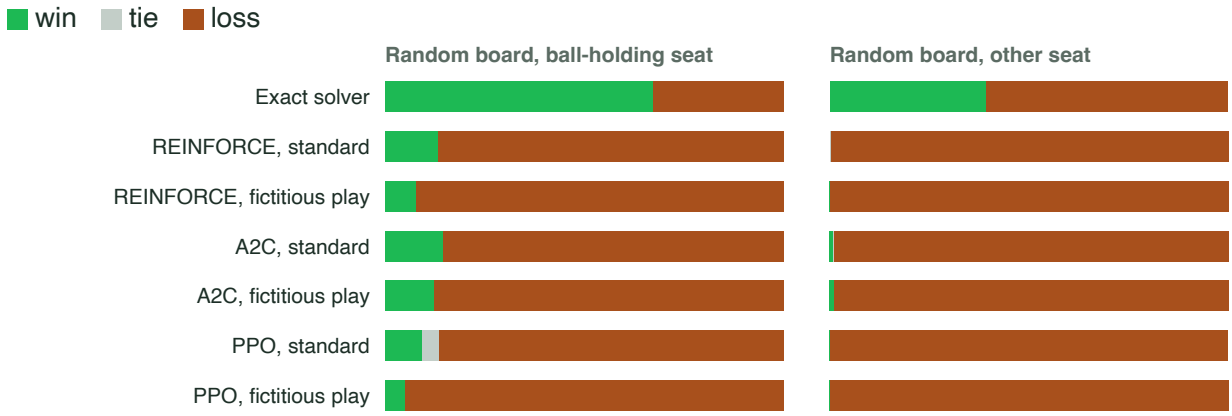
number	what it is
Win rate against the best response	The share of 1,000 games the trained player wins when the opponent plays the exact best response to it, the move that is best against exactly what our player will do. A player who always does the same thing loses every game; the exact solver mixes optimally, so its rate is the ceiling to compare with.
Win rate against the exact Nash policy	The same, against the exact equilibrium player.
Win rate against a random player	A sanity check: every learner should beat a player who moves at random.
A win	The player scores while the opponent does not. On the first two boards the first goal ends the game; in the continuing game play restarts after a goal and a win is more goals than the opponent in 100 steps. Games with no goal, or equal goals, are ties.

Exploitability, the best-response player's gain, is not reported: it measures the attacker, and it is the player we train that we want to judge.

Why the exact solver does not win 0% against the best response

The best response is the move that minimizes our player's *expected discounted score difference*, not the one that minimizes its chance of winning. At the random board's kickoff the player holding the ball is ahead: the exact solution is worth +0.15 to it even against a perfect best response. So the exact solver wins 67% of games and loses 33% from that seat, and wins 39% from the other seat. The first

number is the one reported above (our learners always sit in the ball-holding seat); the second is what it gets from the seat without the ball. On the deterministic board both seats tie every game.

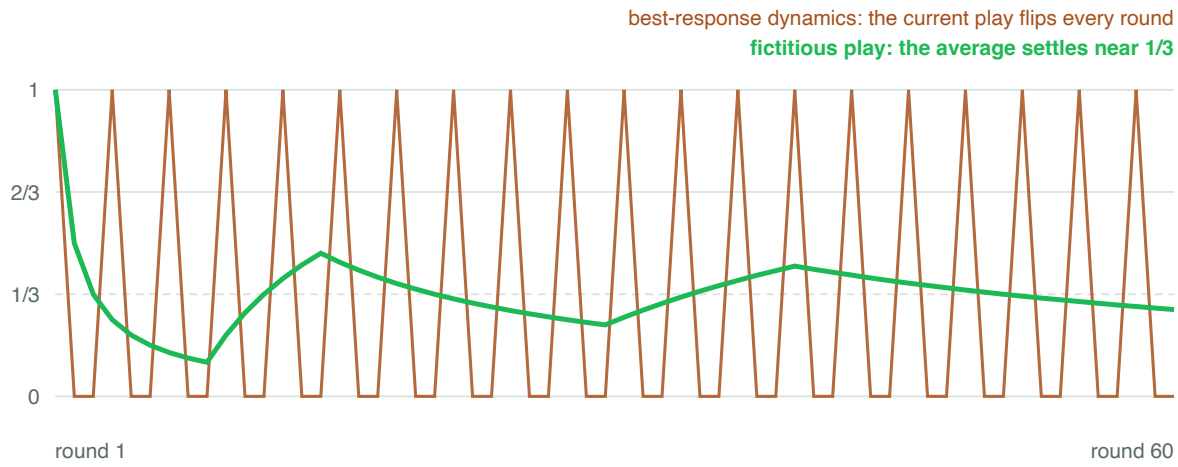


2. How it is implemented

setting	value
Game	7×5 board, goal rows 1–3. Fixed 100 steps, discount 0.9, and the number of steps left is a network input.
Returns	REINFORCE: the discounted return from each step to the end of the episode, no baseline. A2C: 10-step bootstrapped advantage from a learned critic. A2C with the exact critic: the critic is the exact solver's value, frozen, so the advantage is the exact Q(s, a_0, a_1) minus V(s). PPO: GAE (λ 0.95), ratio clipped at 0.2, 4 epochs.
Updates	Every iteration: 64 episodes of 100 steps, then one gradient step per player on all of that data (four for PPO). 2,000 iterations. Learning rate 10^{-3}, entropy bonus 0.01, 64×64 network, softmax output, a separate network for each player. No mini-batches, no replay.
Standard	Both players' current networks play each other and keep updating: real-time best responses.
Fictitious play	Each player trains against a frozen snapshot of the other, drawn at random from all its past snapshots (one every 5 iterations); the policy it reports is the average of its own snapshots. Every snapshot follows a fixed budget of 5 policy-gradient iterations, and what is averaged is policies (action probabilities), never network weights.

3. Why fictitious play

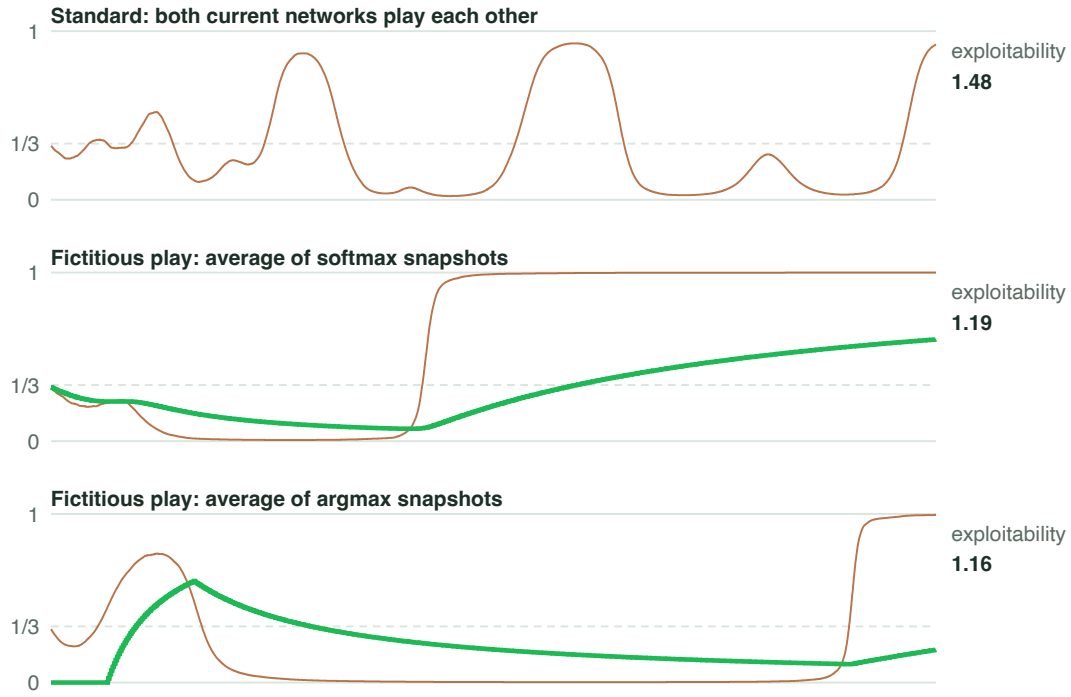
On rock-paper-scissors, answering the opponent's latest move makes the play flip between rock, paper and scissors for ever. Answering the average of everything it has played converges to one third each.



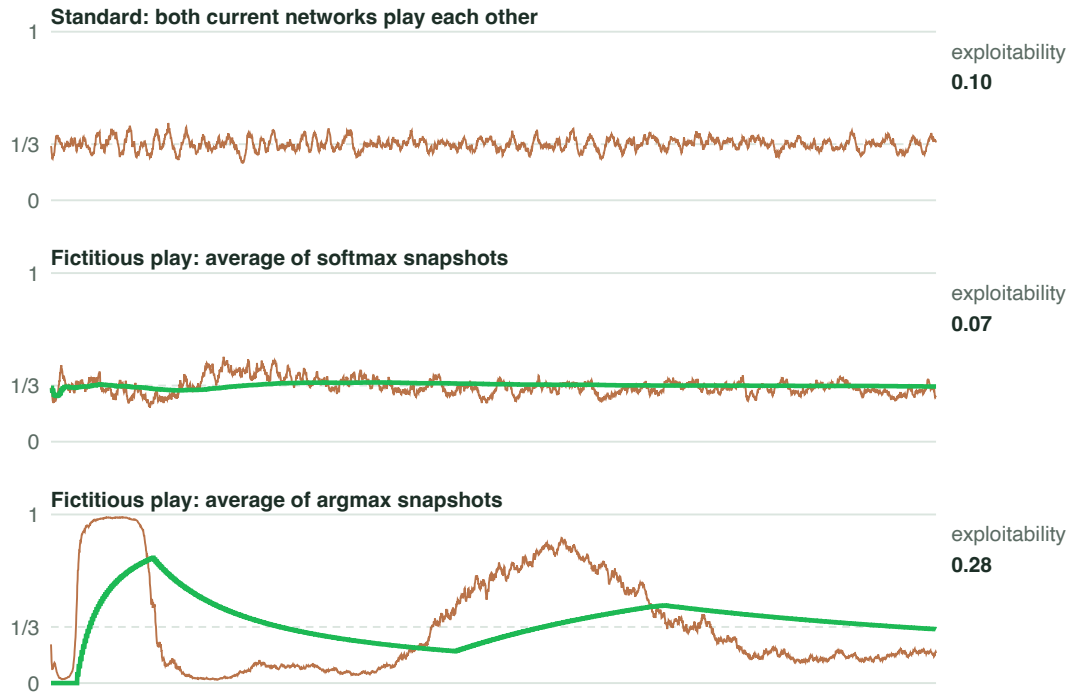
The same test with neural networks

Each player is a small network with a softmax output, trained by REINFORCE on sampled games (3,000 iterations, 64 games each, 3 seeds; the plots show seed 0). **Standard** trains the two current networks against each other. **Fictitious play** trains each player against a snapshot of the other drawn from all its past snapshots and reports the average of a player's snapshots; **argmax** first turns each snapshot into the pure policy that plays its most likely move, so the average is how often the snapshots play each move. Orange: the current network's share of rock; green: the aggregate. Exploitability, on the right, is the mean over seeds and the last 300 iterations, and is 0 at one third each. With the soccer settings the standard networks cycle (exploitability 1.48), and neither form of fictitious play reaches one third (softmax 1.19, argmax 1.16). With the larger entropy bonus all three settle close to the equilibrium (standard 0.10, softmax 0.07, argmax 0.28): this setting damps the cycling, so the standard networks settle too (a stabilization effect, not evidence of convergence to the unregularized equilibrium).

Soccer settings: Adam at 0.001, entropy bonus 0.01



Larger entropy bonus: plain gradient steps at 0.1, entropy bonus 0.2



4. Results

Win rates against the best response, by kickoff seat

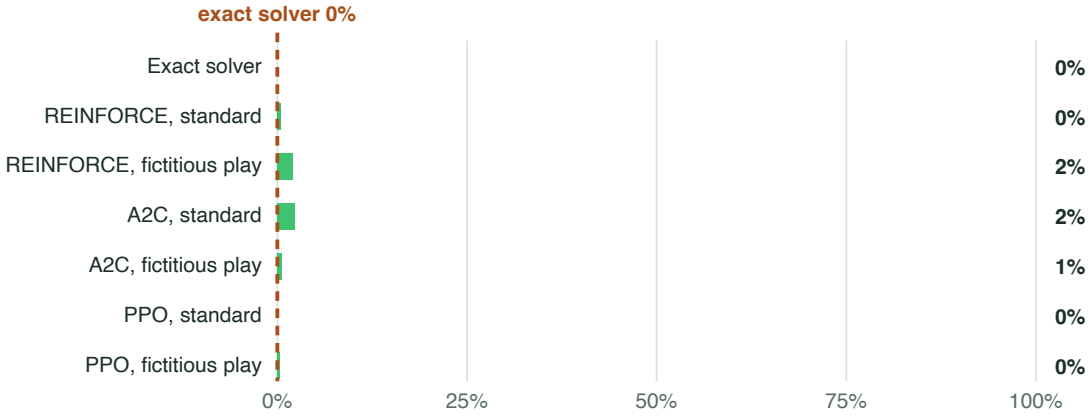
The share of 1,000 games each learner wins when the opponent plays the exact best response to it (the move that minimizes its expected discounted score difference). **Ball seat**: the player that holds the ball at kickoff. **Other seat**: the same learner as the other player. **Balanced**: the 50/50 average of the two, one number that does not depend on who starts with the ball. Mean of 3 seeds; the exact solver is the top row, as context rather than a target. The deterministic board is the A10 board: the exact solver ties itself there, so it wins 0% from both seats. A dash means the learner was not run on that board. Wins against the exact Nash policy and a random player are in the charts below.

learner	The deterministic board (A10)			The random move-order board			The continuing game		
	ball seat	other seat	balanced	ball seat	other seat	balanced	ball seat	other seat	balanced
Exact solver	**0%**	**0%**	**0%**	**67%**	**39%**	**53%**	**0%**	**0%**	**0%**
REINFORCE, standard	0%	1%	0%	13%	0%	7%	0%	0%	0%
REINFORCE, fictitious play	2%	1%	2%	8%	0%	4%	0%	0%	0%
A2C, standard	2%	1%	2%	14%	1%	8%	0%	0%	0%
A2C, fictitious play	1%	2%	1%	12%	1%	7%	0%	0%	0%
PPO, standard	0%	0%	0%	9%	0%	5%	0%	0%	0%
PPO, fictitious play	0%	0%	0%	5%	0%	2%	0%	0%	0%
A2C, exact critic, standard	–	–	–	–	–	–	0%	0%	0%
A2C, exact critic, fictitious play	–	–	–	–	–	–	0%	0%	0%

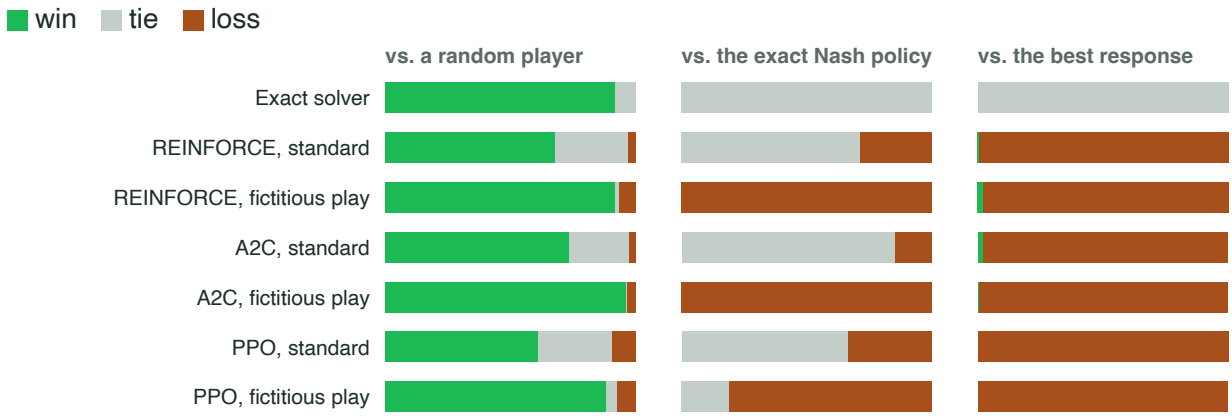
The deterministic board (A10)

Players move simultaneously and the carrier wins every contested square, so the exact solution is pure everywhere and ties itself.

Win rate against the best response (mean of 3 seeds; dashed line: the exact solver). The exact solver wins 0%; the learners win 0% to 2%.



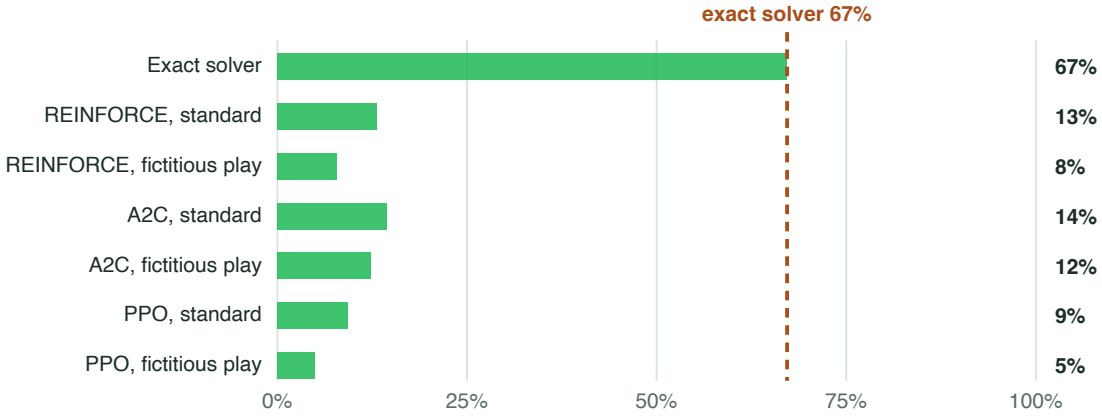
Wins, ties and losses against each opponent. Against the exact Nash policy the standard learners tie 67% to 85% of games and the fictitious-play learners 0% to 19%.



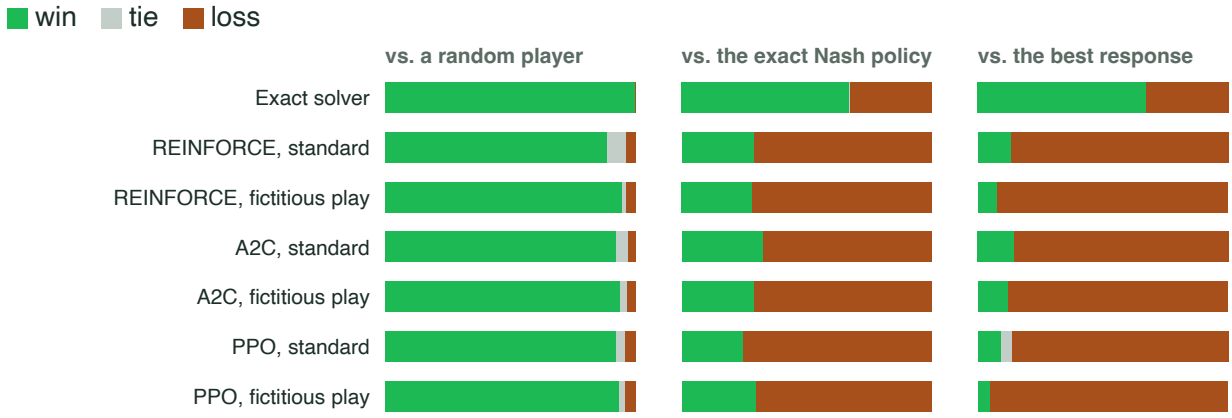
The random move-order board

Moves are applied in a random order each step; the exact solution mixes at 94 stage games.

Win rate against the best response (mean of 3 seeds; dashed line: the exact solver). The exact solver wins 67%; the learners win 5% to 14%.



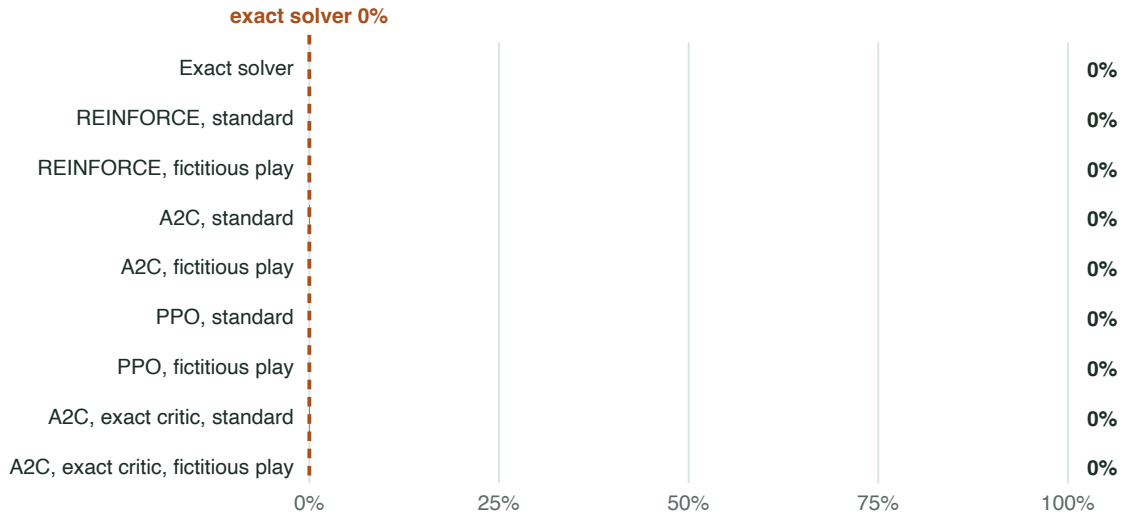
Wins, ties and losses against each opponent.



The continuing game

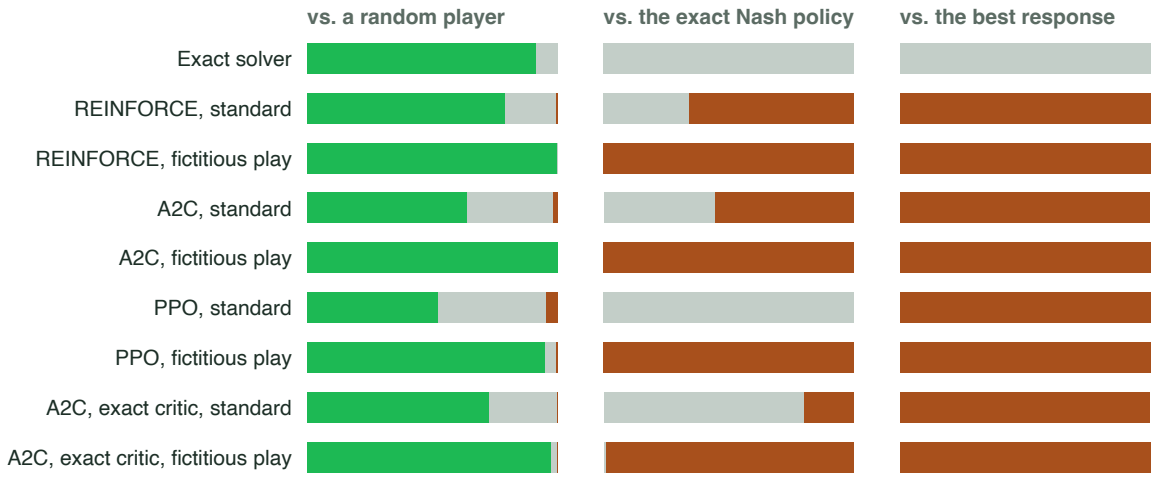
Play restarts after every goal and runs for exactly 100 steps; a win is more goals than the opponent. Also shows A2C with the exact solver's values as a frozen critic.

Win rate against the best response (mean of 3 seeds; dashed line: the exact solver). The exact solver wins 0%; the learners win 0% to 0%.



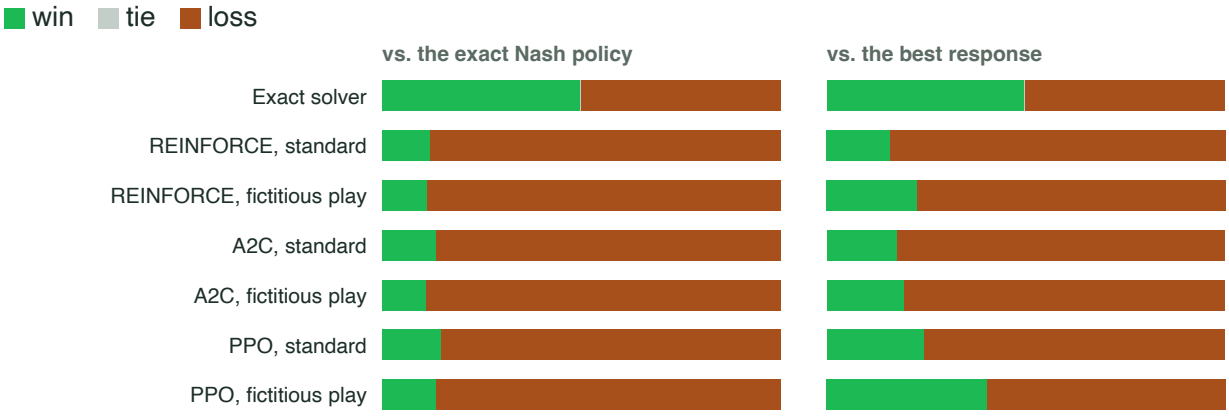
Wins, ties and losses against each opponent. Against the exact Nash policy the standard learners tie 34% to 100% of games and the fictitious-play learners 0% to 1%. A2C with the exact critic ties the exact Nash policy in 80% of games, against 45% for A2C with a learned critic.





5. Mixed states of the random board

At 94 states the exact equilibrium has to randomize. Each learner starts 20 games from every one of them. Distance is the total-variation distance between the learner's probabilities and the exact mix at the first step (0 = identical, 1 = no overlap); "player 0 mixes" is the share of states where it puts less than 90% on its top move.

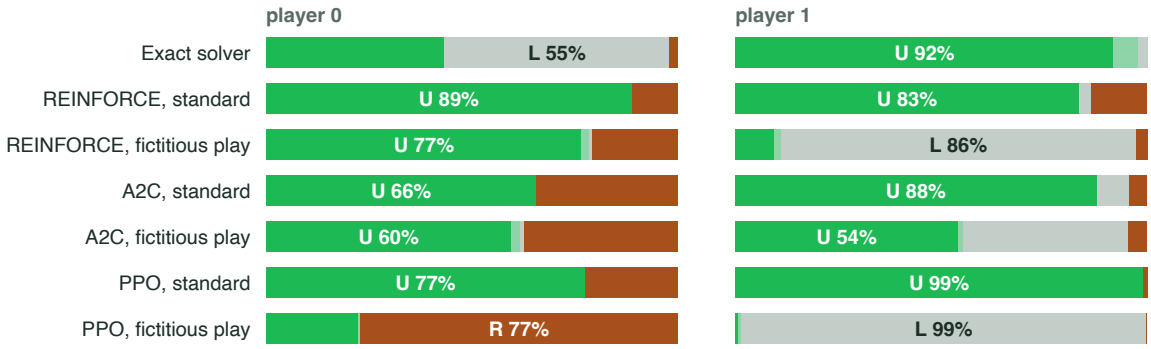


The distance from the exact mix is 0.59 to 0.67 for every learner. Player 0 mixes (puts under 90% on its top move) at 29% to 100% of these states, though not at the states or in the proportions the exact solution does.

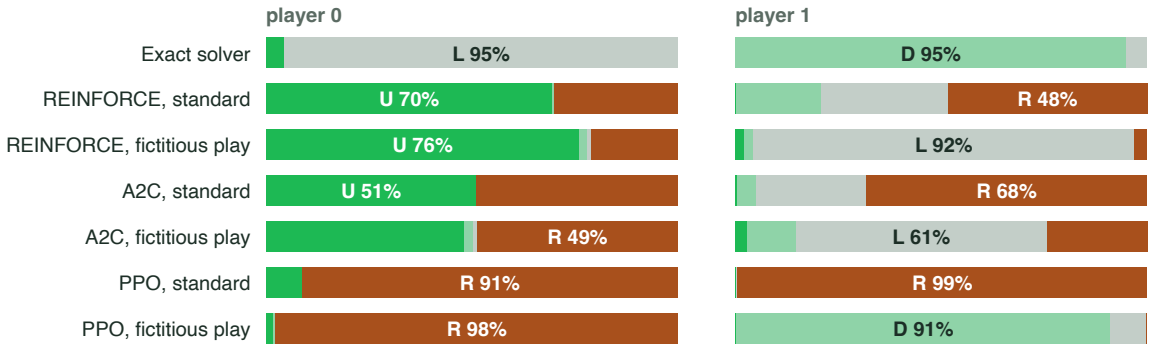
Three of these states in full

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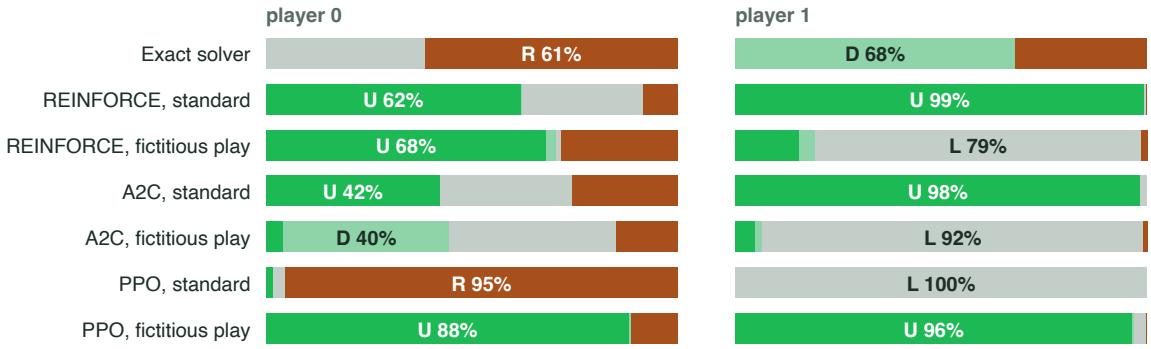
Mixed state 1 (three-way mix): player 0 at (0, 0), player 1 at (2, 0), player 0 has the ball



Mixed state 2 (two-way mix): player 0 at (0, 0), player 1 at (1, 1), player 0 has the ball



Mixed state 3 (two-way mix): player 0 at (0, 1), player 1 at (1, 0), player 1 has the ball

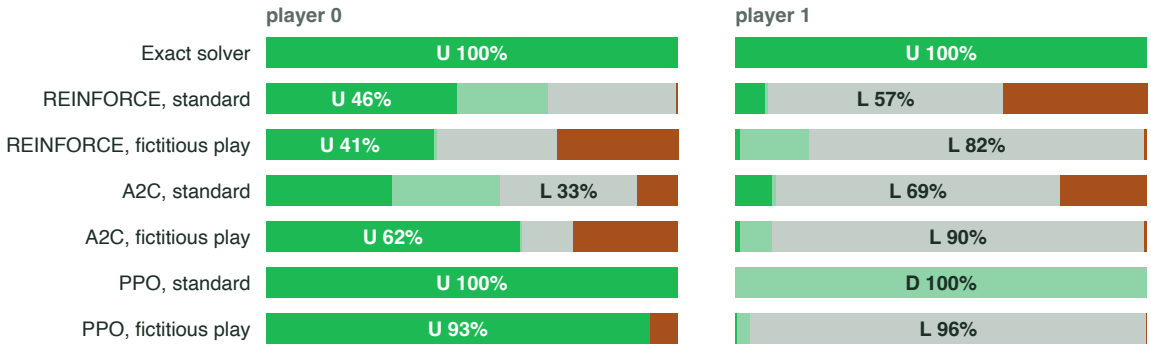


6. Action probabilities

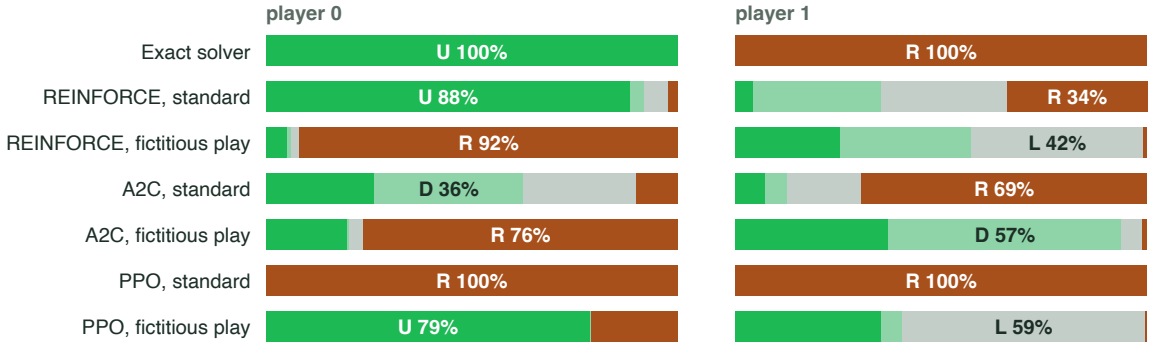
The probability each trained network gives to every move, at four fixed positions of the deterministic board (seed 0), next to the exact solver's move.

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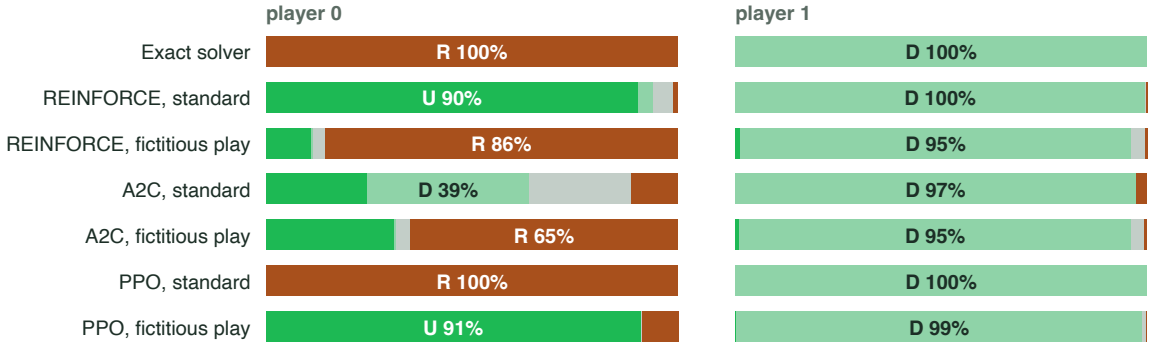
Kickoff: player 0 at (0, 1), player 1 at (6, 3), player 0 has the ball



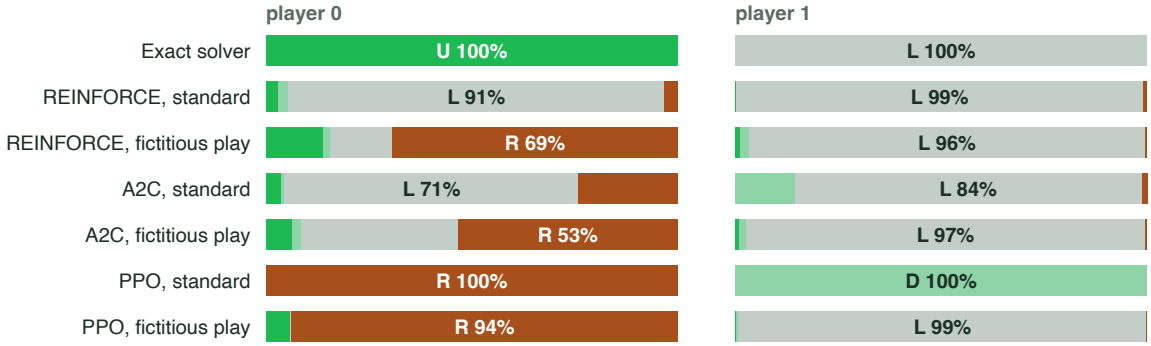
Players side by side in a goal row: player 0 at (3, 2), player 1 at (4, 2), player 0 has the ball



Carrier near the goal, defender far away: player 0 at (5, 1), player 1 at (6, 3), player 0 has the ball

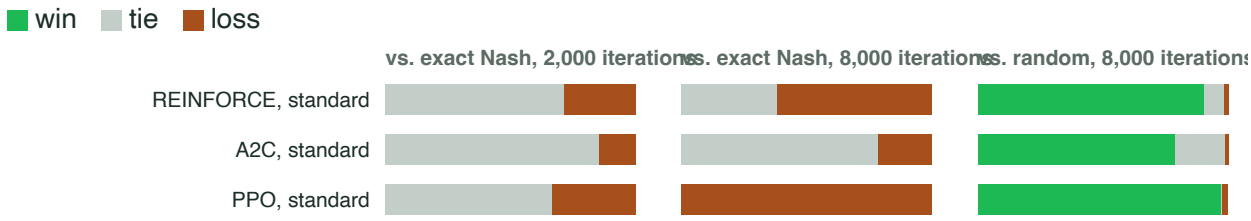


The right player holds the ball, players stacked: player 0 at (3, 2), player 1 at (3, 3), player 1 has the ball



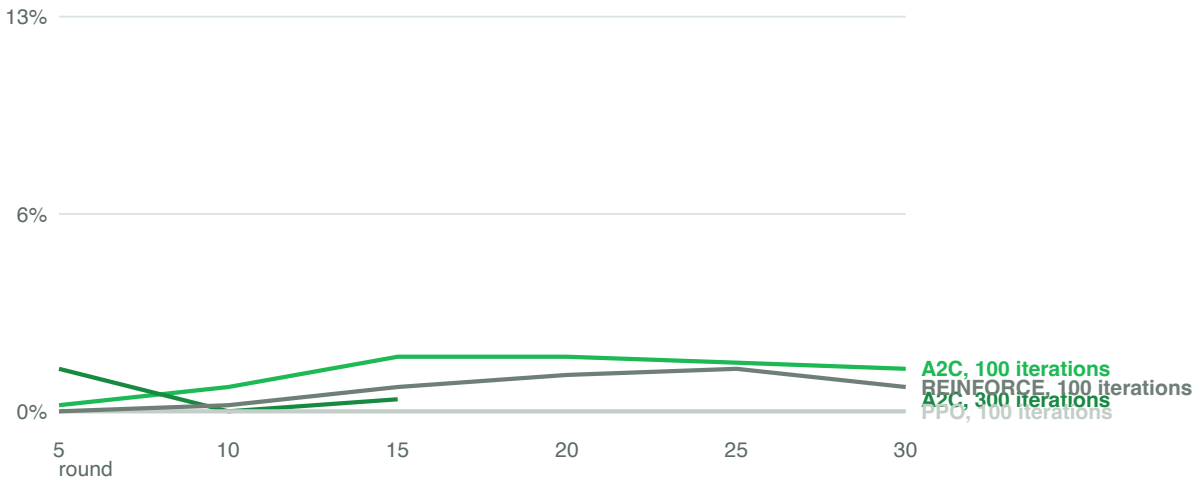
7. More training

Standard training at 8,000 instead of 2,000 iterations (two seeds each), deterministic board.



8. Fictitious play with long best responses

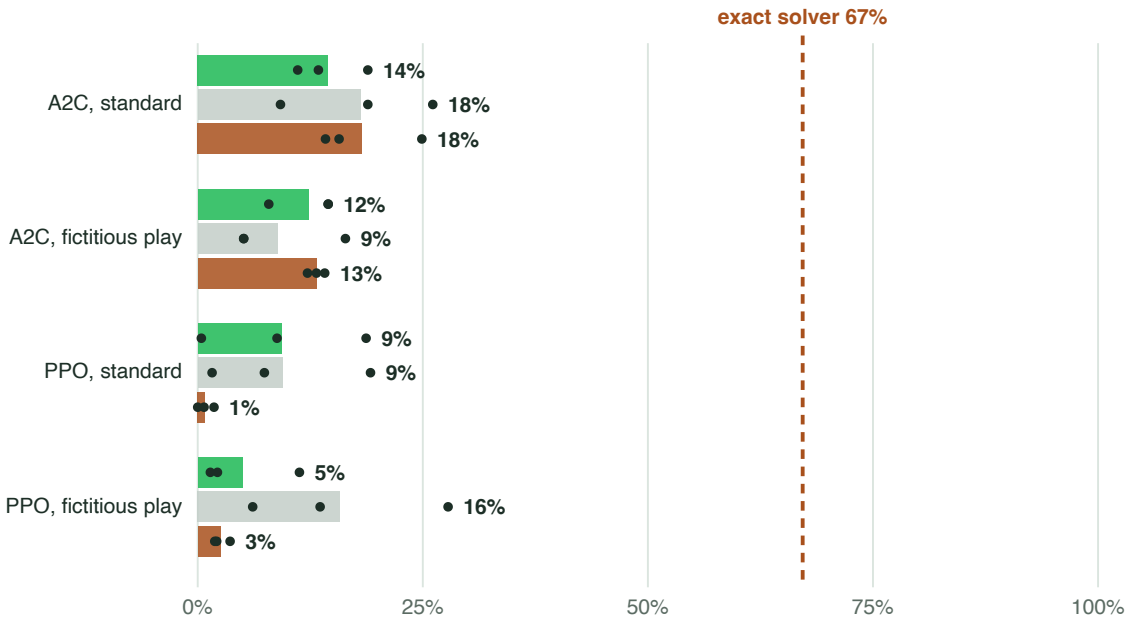
Each round a player trains for 100 (or 300) iterations against the average of its opponent's earlier best responses, then adds the result to its history. Win rate against the best response at each checkpoint round (one seed per run); every line ends below 2%.



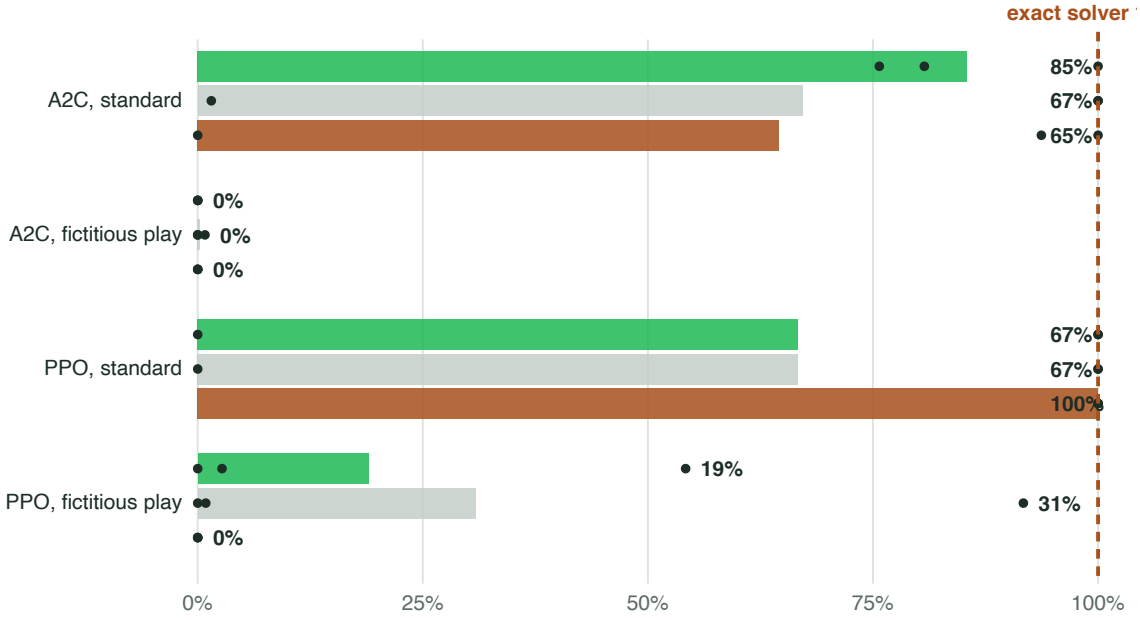
9. Network sharing and trimming

A2C and PPO, three seeds each, on all three boards. **Shared network**: both players' policies are two output slices of one network, so each player's update also moves the other. **Last 10 steps left out**: the final 10 of the 100 steps of every episode are not used in the loss, though they still count in the returns of earlier steps. Bars are the mean over seeds; dots are single seeds. ■ separate networks, ■ one shared network, ■ last 10 steps left out.

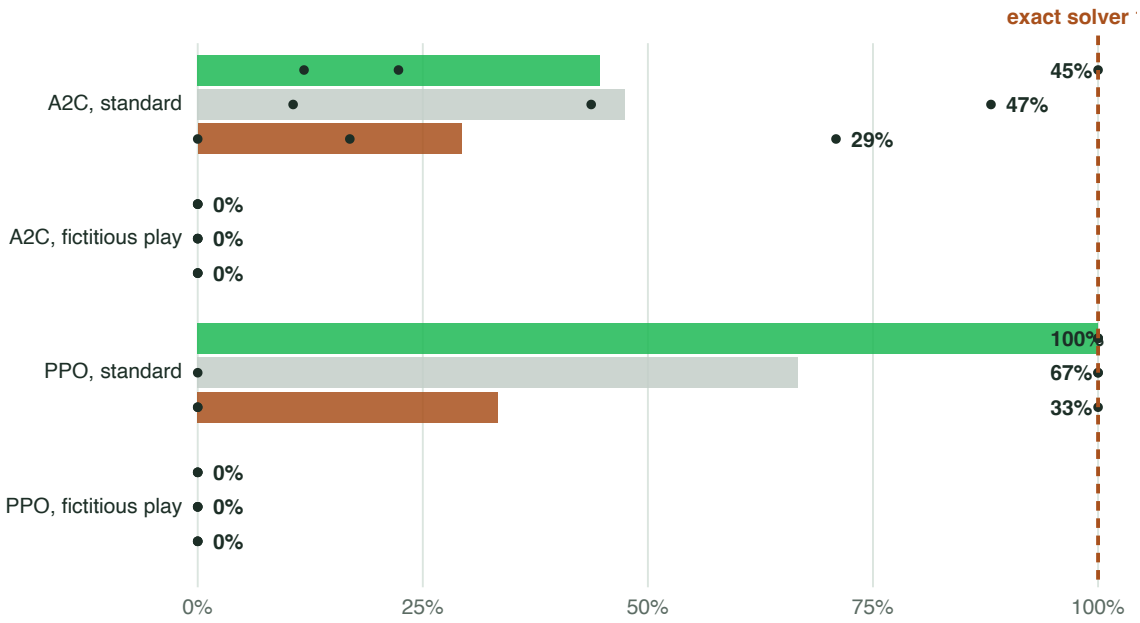
Random move-order board: win rate against the best response



Deterministic board: games tied with the exact Nash policy



Continuing game: games tied with the exact Nash policy



Random board: against separate networks, a shared network changes the win rate by +4, –3, +0, +11 points (A2C standard, A2C fictitious play, PPO standard, PPO fictitious play); single seeds of the same setup differ by up to 22 points. Leaving out the last 10 steps changes A2C by +4 and +1 points, but PPO with standard training stops scoring: it wins 19% of games against a random player, against 92% before. No variant gets close to the exact solver.

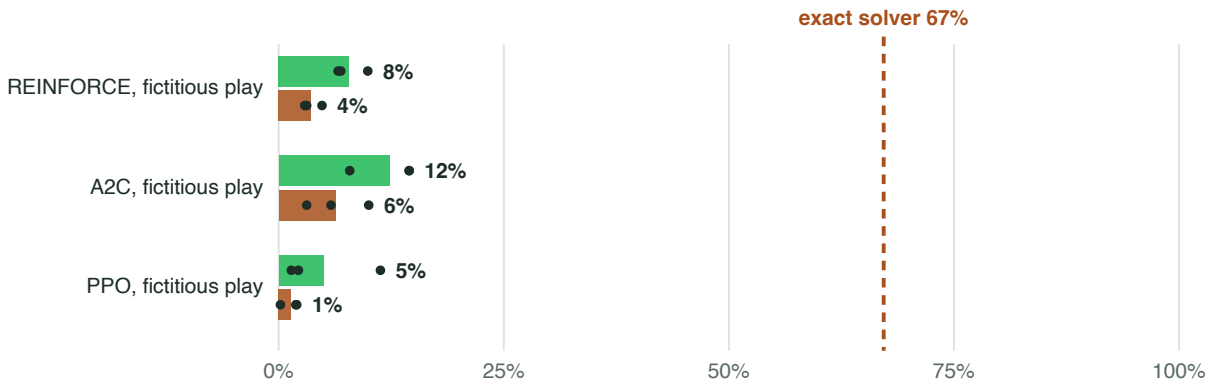
Deterministic board: no variant wins more than 2% of games against the best response, as for the baseline (the exact solver wins 0%, since it ties itself). The share of games tied with the exact Nash policy changes by –18, +0, +0, +12 points with a shared network and by –21, +0, +33, –19 points with the last 10 steps left out (A2C standard, A2C fictitious play, PPO standard, PPO fictitious play); single seeds of the same setup differ by up to 100 points. A tie does not mean equilibrium play: PPO with standard training ties all games once the last 10 steps are left out, but wins only 36% of games against a random player, against 61% before.

Continuing game: no variant wins more than 0% of games against the best response, as for the baseline. The share of games tied with the exact Nash policy changes by +3, +0, –33, +0 points with a shared network and by –15, +0, –67, +0 points with the last 10 steps left out (A2C standard, A2C fictitious play, PPO standard, PPO fictitious play); single seeds of the same setup differ by up to 100 points, so three seeds cannot settle changes of this size.

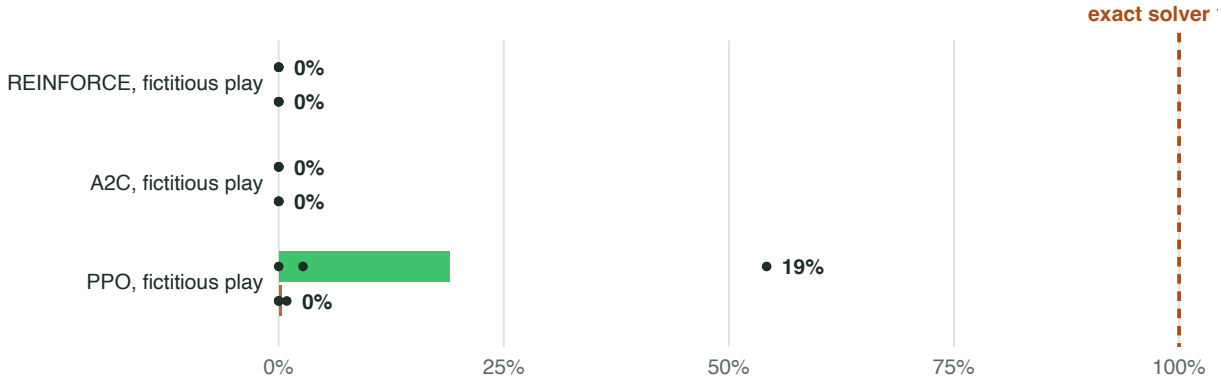
10. Averaging pure policies

Neural networks cannot output a mixed strategy except through their softmax, so each snapshot is made a pure policy (the move with the highest probability), those are averaged, and each player best-responds to that average. This is fictitious play in its original form: each snapshot is a pure best response, and the mix is how often each move was played. Both the opponent that a player trains against and the policy it reports are averages of pure snapshots. Bars are the mean over 3 seeds and dots single seeds; ■ average of softmax snapshots (the fictitious play used elsewhere), ■ average of argmax snapshots.

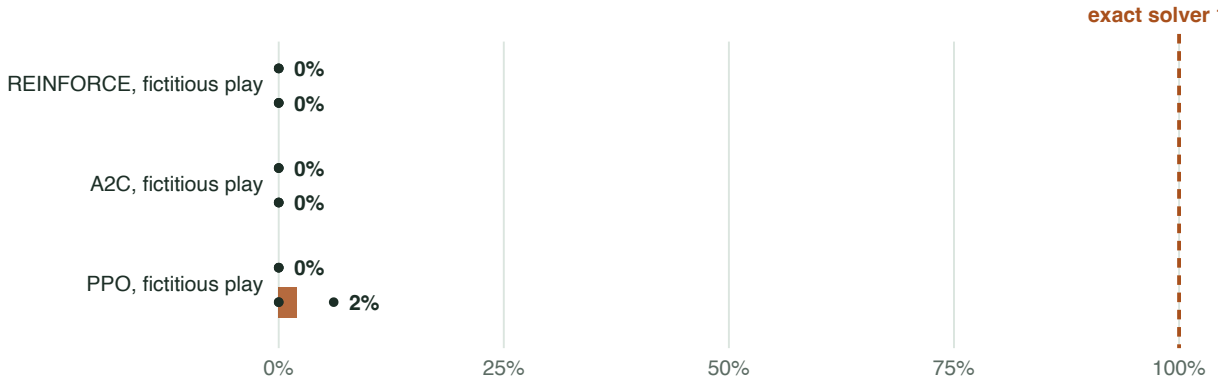
Random move-order board: win rate against the best response



Deterministic board: games tied with the exact Nash policy



Continuing game: games tied with the exact Nash policy



Random board: averaging argmax instead of softmax snapshots changes the win rate against the best response by –4, –6, –4 points (REINFORCE, A2C, PPO); single seeds differ by up to 10 points. Deterministic board: it changes the share of games tied with the exact Nash policy by +0, +0, –19 points; single seeds differ by up to 54 points. Continuing game: the share tied with the exact Nash policy changes by +0, +0, +2 points; single seeds differ by up to 6 points, and no argmax learner wins more than 0% of games against the best response.

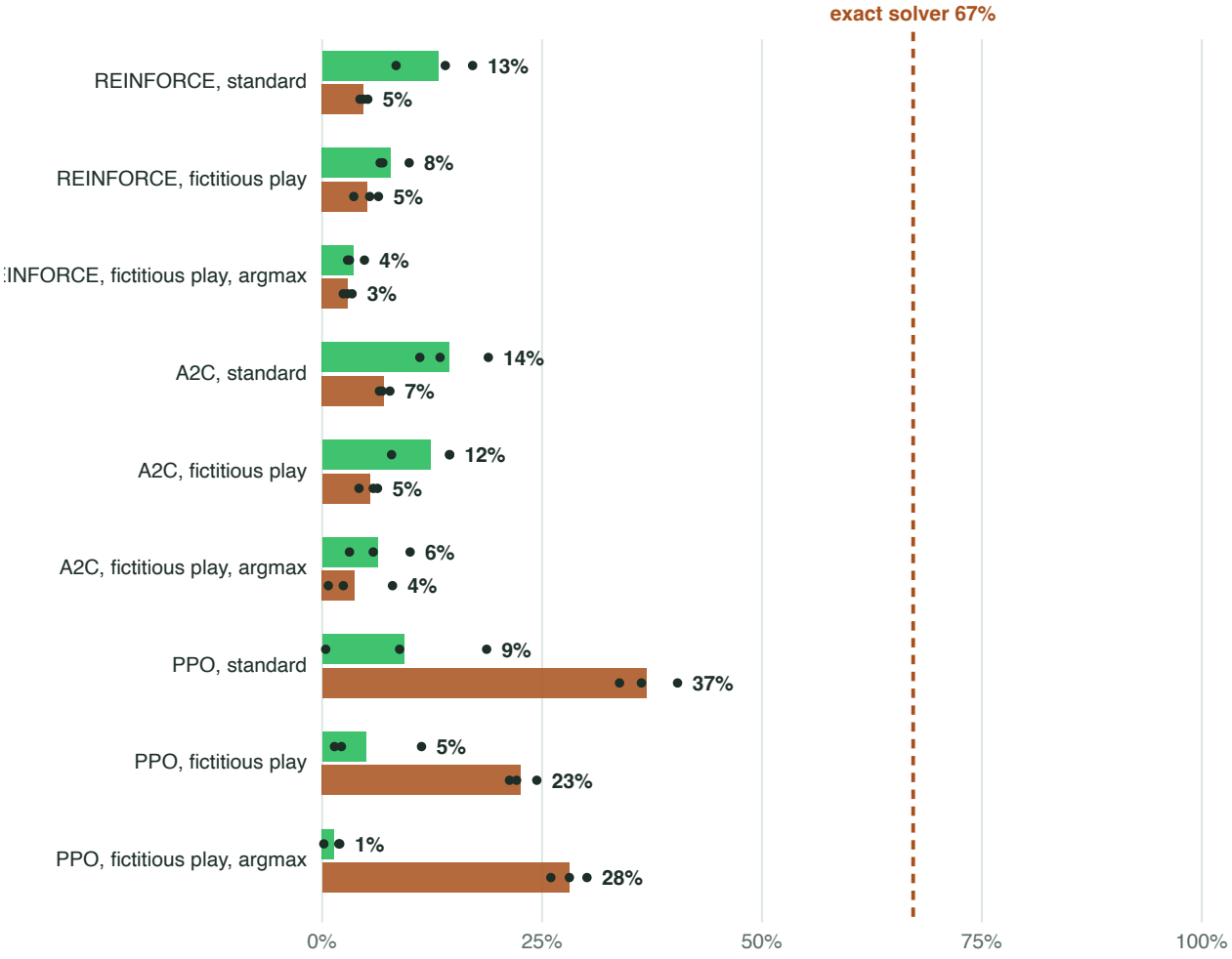
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11. A larger entropy bonus

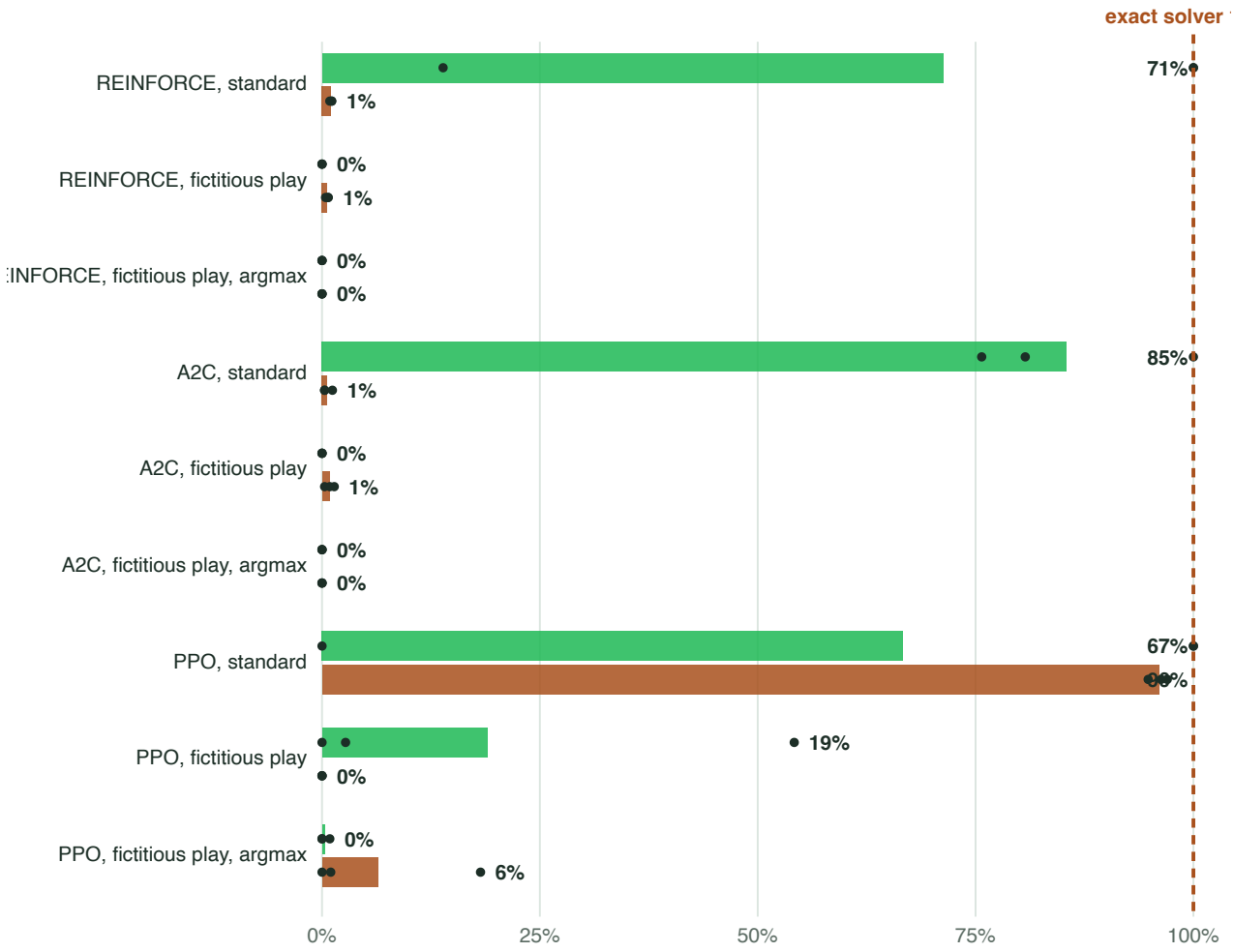
On rock-paper-scissors a larger entropy bonus made the networks settle, but entropy regularization changes the optimization problem, so this is a stabilization heuristic reported next to the original setting, not a way to recover the unregularized equilibrium. Here every learner (REINFORCE, A2C,

PPO), every way of training (standard, fictitious play, fictitious play with argmax snapshots) and every board is trained again with the entropy bonus raised from 0.01 to 0.2; nothing else changes. Bars are the mean over 3 seeds and dots single seeds; ■ entropy bonus 0.01 (used elsewhere), ■ entropy bonus 0.2.

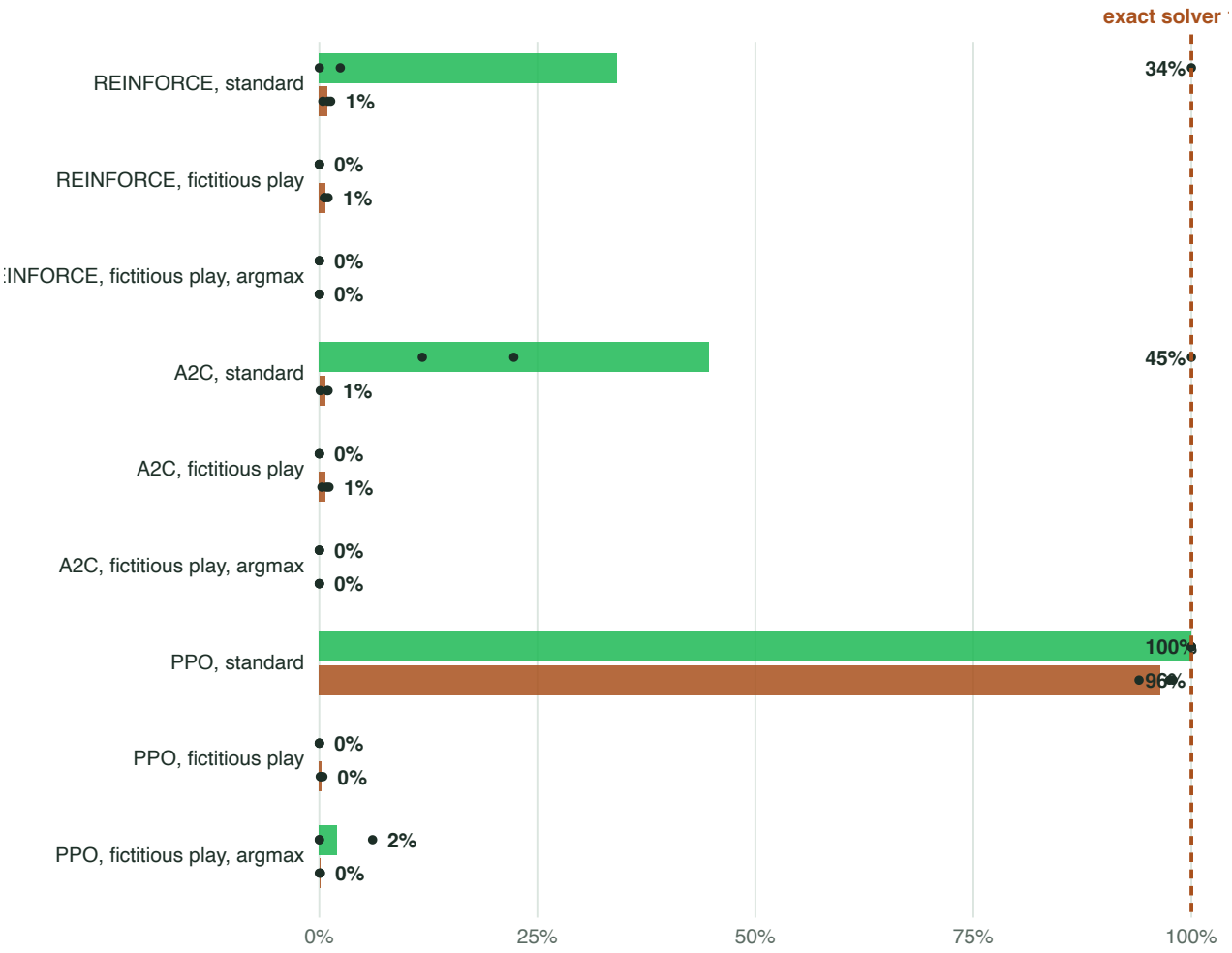
Random move-order board: win rate against the best response



Deterministic board: games tied with the exact Nash policy



Continuing game: games tied with the exact Nash policy



On the random board, averaged over the three ways of training, the larger bonus moves the win rate against the best response by REINFORCE −4, A2C −6, PPO +24 points; the exact solver wins 67% from the ball seat. This is an ablation and does not show convergence to the unregularized equilibrium.

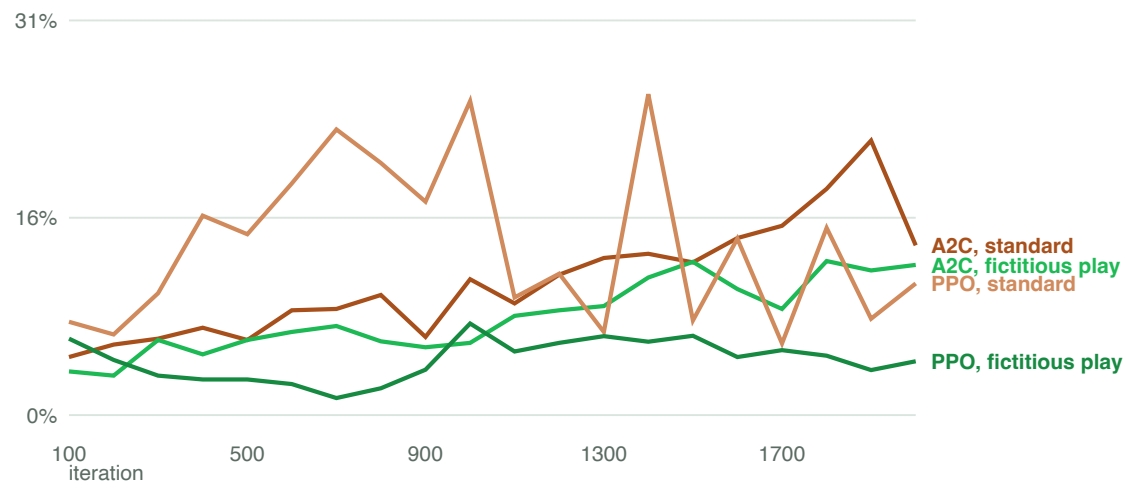
Random board: the larger bonus changes the win rate against the best response by −8, −3, −1, −7, −7, −3, +28, +18, +27 points; against a random player the win rate changes by +1, −8, +5, −2, −6, +4, +6, +4, +3 points (REINFORCE, A2C, PPO; standard, fictitious play, argmax). Deterministic board: the larger bonus changes the share tied with the exact Nash policy by −70, +1, +0, −85, +1, +0, +29, −19, +6 points; against a random player the win rate changes by +22, −2, +5, +19, −6, +4, +32, +11, +13 points (REINFORCE, A2C, PPO; standard, fictitious play, argmax). Continuing game: the larger bonus changes the share tied with the exact Nash policy by −33, +1, +0, −44, +1, +0, −4, +0, −2 points; against a random player the win rate changes by +6, −17, +6, +24, −16, +6, +43, +5, +11 points (REINFORCE, A2C, PPO; standard, fictitious play, argmax).

12. Does training converge?

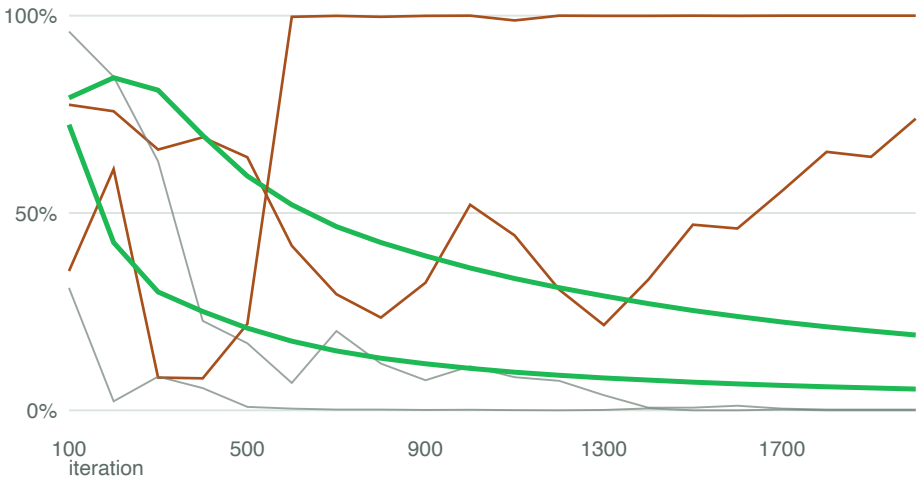
Best-response dynamics (standard training) can cycle; fictitious play averages the past. Every 100 iterations the policy each learner would report is scored against the exact best response (mean of 3 seeds), and the probability it gives to the exact solver's kickoff move is recorded for seed 0: the current network (thin) and, for fictitious play, the average (thick). Orange: standard training's current

network. Grey: the current network under fictitious play. Green: fictitious play's average. A2C and PPO are drawn together in each chart.

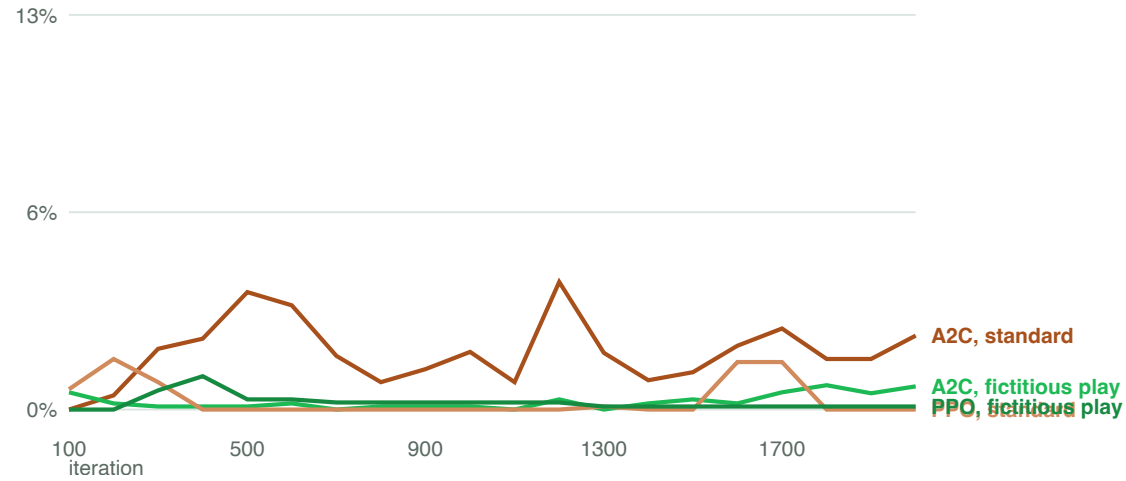
Random move-order board: win rate against the best response



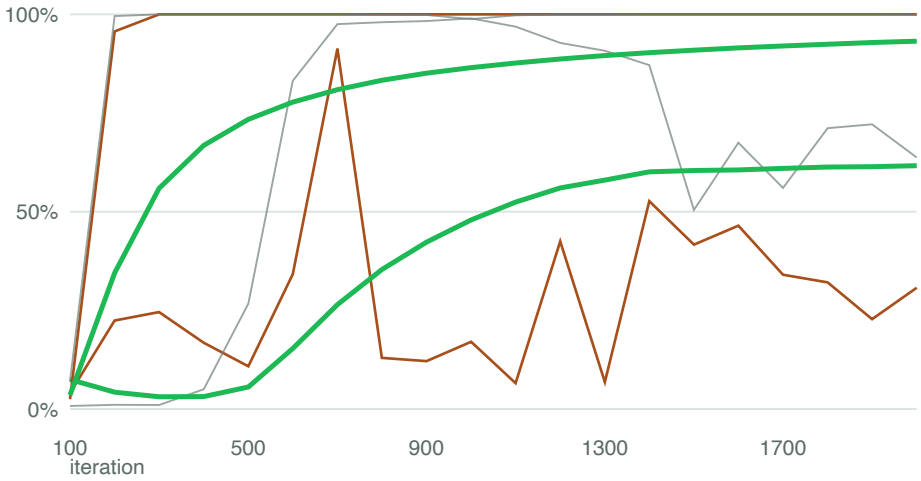
Random move-order board: probability of the exact solver's kickoff move (R; exact solver 100%)



Deterministic board: win rate against the best response



Deterministic board: probability of the exact solver's kickoff move (U; exact solver 100%)



Random board, second half of training: the win rate against the best response stays within A2C standard 9% to 22%; A2C fictitious play 8% to 12%; PPO standard 6% to 25%; PPO fictitious play 4% to 6%. The exact solver wins 67% from the ball seat.

Deterministic board, second half of training: the win rate against the best response stays within A2C standard 1% to 4%; A2C fictitious play 0% to 1%; PPO standard 0% to 2%; PPO fictitious play 0% to 0%. The exact solver wins 0% from the ball seat.

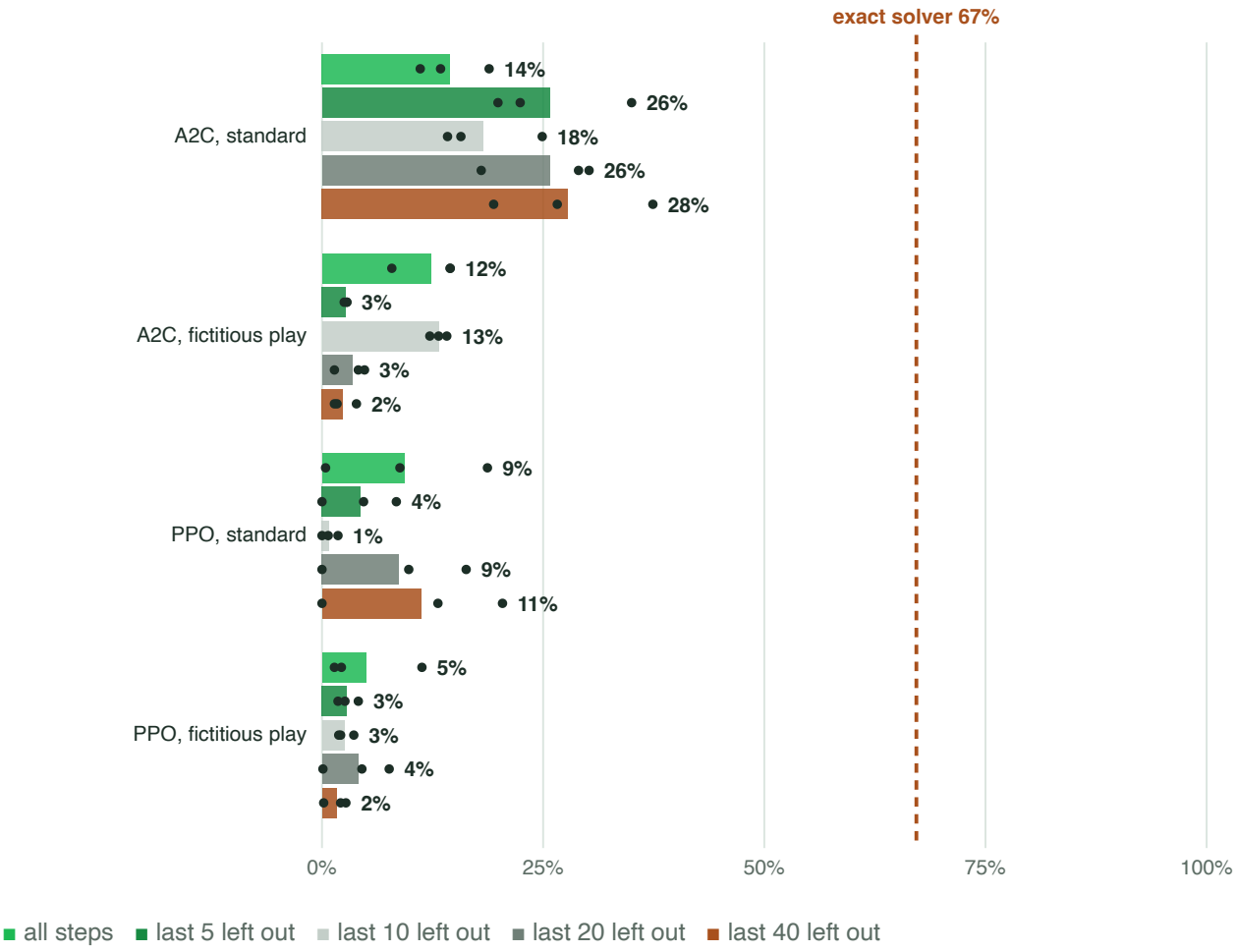
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13. Training choices

Each comparison changes one thing from the usual setting (2,000 iterations of 64 episodes, all data used once, a snapshot every 5 iterations, all 100 steps in the loss) and keeps the rest. Bars are the mean over 3 seeds and dots single seeds.

Leaving out the last steps: how many?

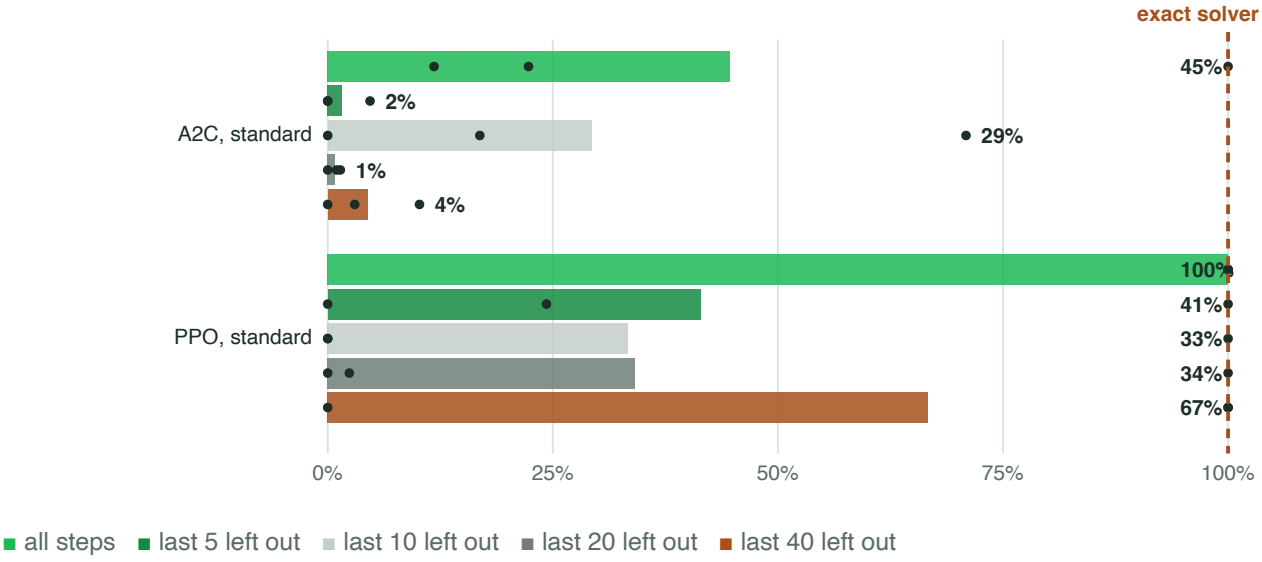
Random move-order board; the last N of the 100 steps of every episode are left out of the loss.



Change in the win rate against the best response, in points, from "all steps" (rows in the order shown): last 5 left out: +11, −10, −5, −2; last 10 left out: +4, +1, −8, −2; last 20 left out: +11, −9, −1, −1; last 40 left out: +13, −10, +2, −3. Single seeds of one setting differ by up to 20 points.

Leaving out the last steps in the continuing game

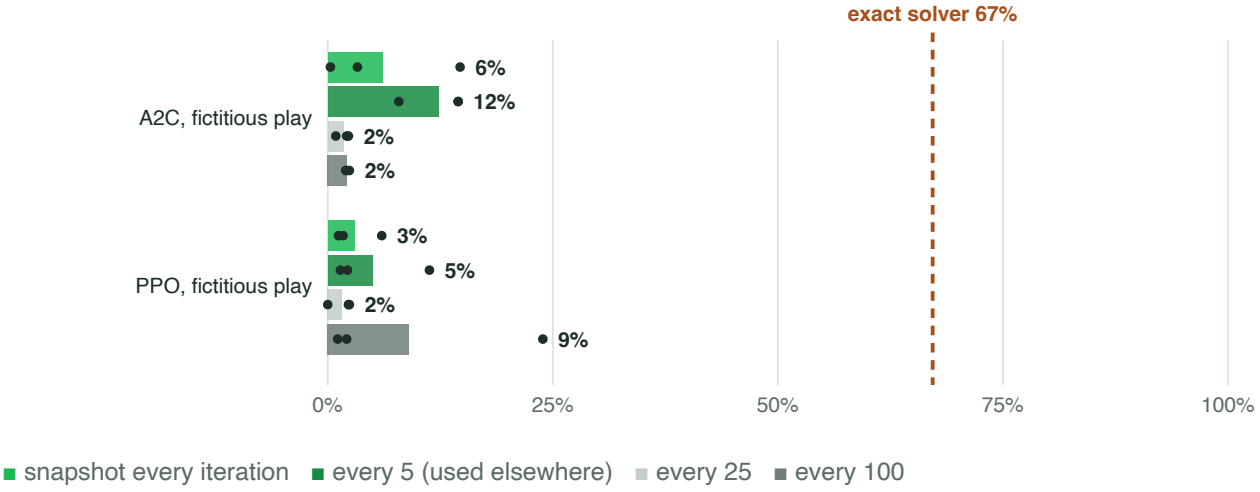
Continuing game, standard training; share of games tied with the exact Nash policy.



Change in the share tied with the exact Nash policy, in points, from "all steps" (rows in the order shown): last 5 left out: −43, −59; last 10 left out: −15, −67; last 20 left out: −44, −66; last 40 left out: −40, −33. Single seeds of one setting differ by up to 100 points.

How long to train between snapshots

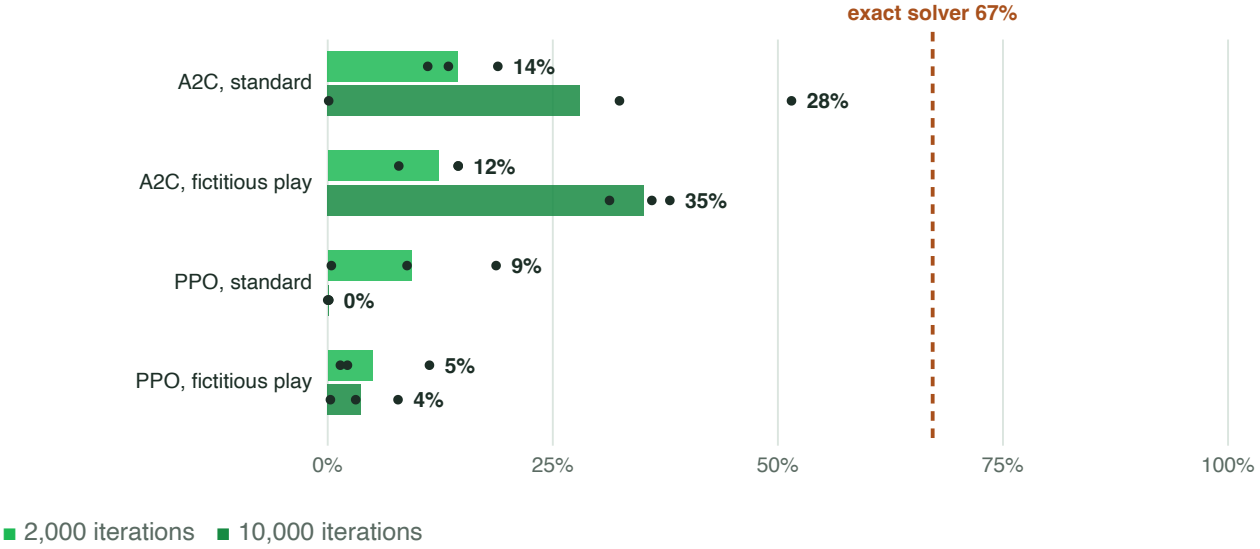
Random board, fictitious play, 2,000 iterations in total: the training budget of each best response.



Change in the win rate against the best response, in points, from "snapshot every iteration" (rows in the order shown): every 5 (used elsewhere): +6, +2; every 25: –4, –1; every 100: –4, +6. Single seeds of one setting differ by up to 23 points.

Five times as many iterations

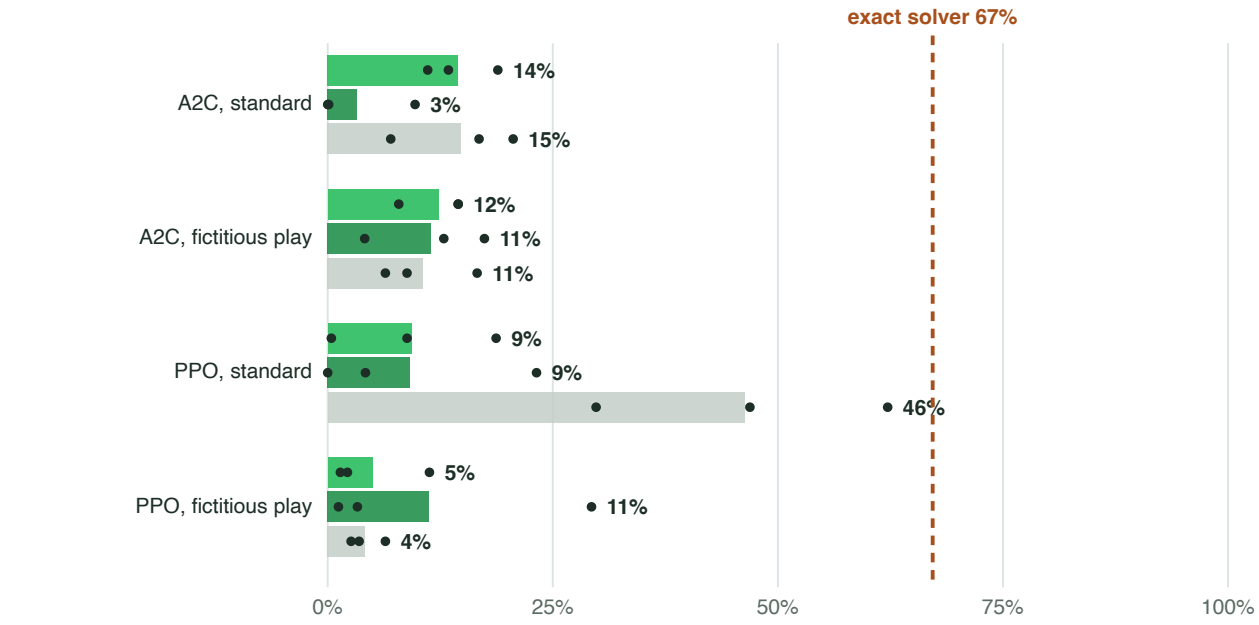
Random board: 10,000 iterations instead of 2,000.



Change in the win rate against the best response, in points, from "2,000 iterations" (rows in the order shown): 10,000 iterations: +14, +23, –9, –1. Single seeds of one setting differ by up to 51 points.

Mini-batches and replayed data

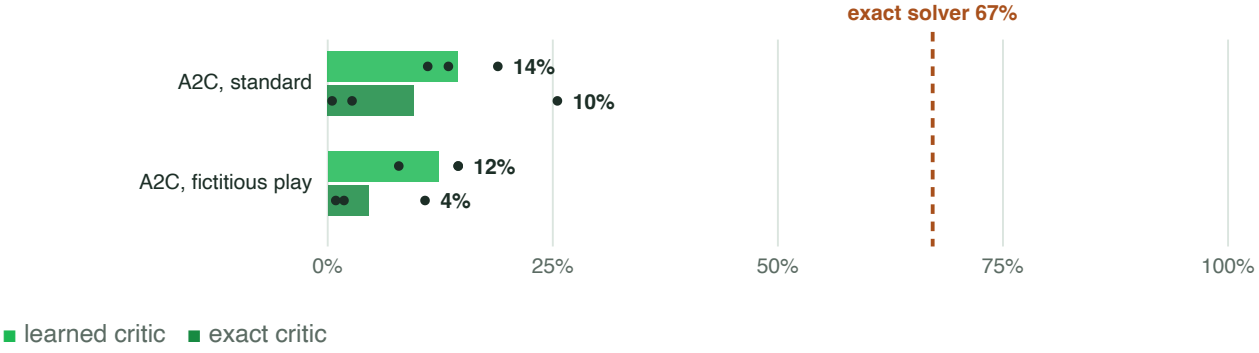
Random board. Mini-batch: many gradient steps per iteration, each on 256 random samples. Replay: each update also uses the data of the 3 previous iterations.



Change in the win rate against the best response, in points, from "all data, one step" (rows in the order shown): mini-batches of 256: –11, –1, –0, +6; replay of 4 iterations: +0, –2, +37, –1. Single seeds of one setting differ by up to 32 points.

The exact solver's values as the critic: random board

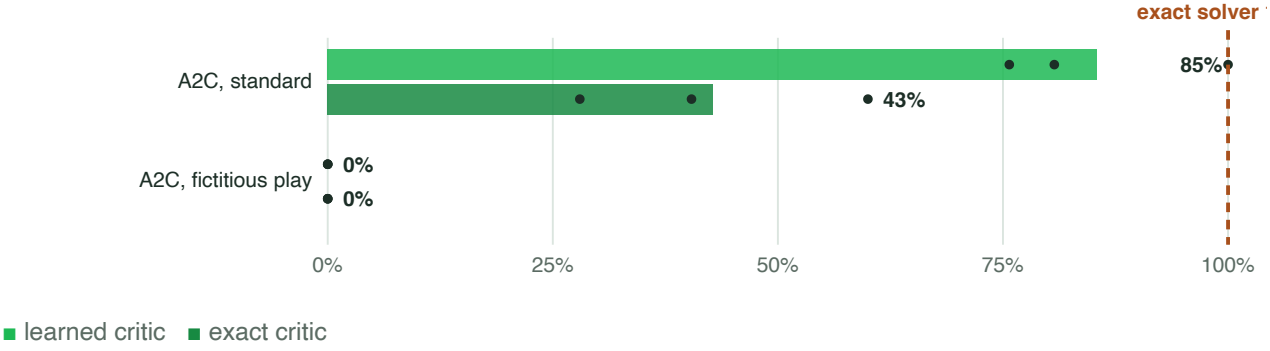
A2C with a learned critic against A2C with the exact solver's values as a frozen critic.



Change in the win rate against the best response, in points, from "learned critic" (rows in the order shown): exact critic: –5, –8. Single seeds of one setting differ by up to 25 points.

The exact solver's values as the critic: deterministic board

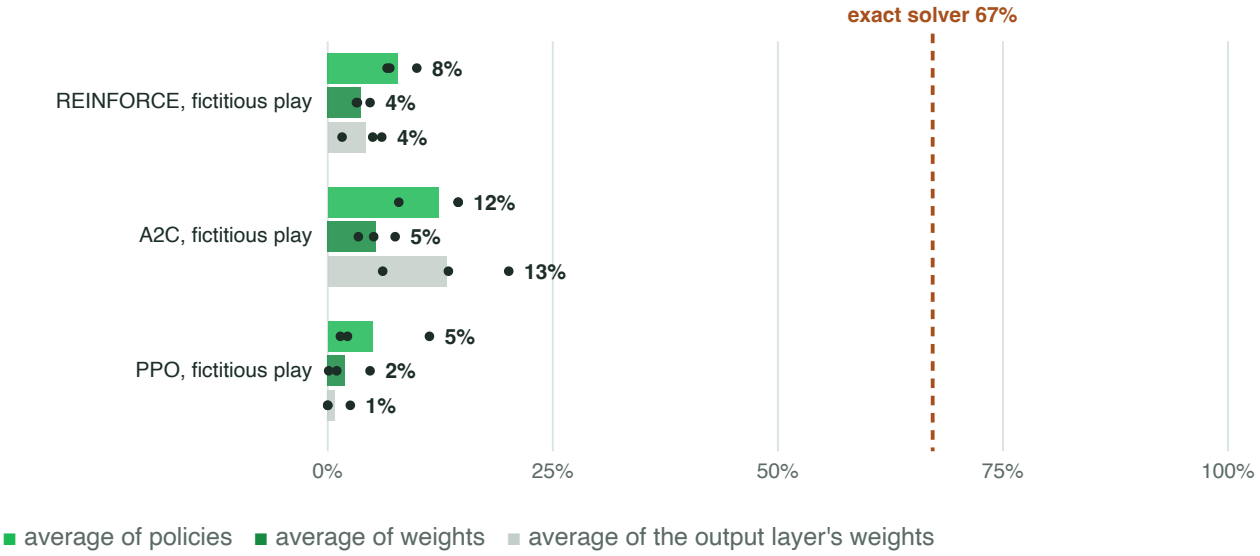
A2C with a learned critic against A2C with the exact solver's values as a frozen critic.



Change in the share tied with the exact Nash policy, in points, from "learned critic" (rows in the order shown): exact critic: –43, +0. Single seeds of one setting differ by up to 32 points.

Averaging weights instead of policies

Random board, fictitious play. Fictitious play averages policies; the other two build one network from the snapshots' weights (all layers, or only the output layer).



Change in the win rate against the best response, in points, from "average of policies" (rows in the order shown): average of weights: –4, –7, –3; average of the output layer's weights: –4, +1, –4. Single seeds of one setting differ by up to 14 points.

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14. Related work

Has fictitious play been combined with policy gradient before? Yes, in several forms. In a short search I found these; the one-line summaries follow each paper's abstract or listing and I have not checked the proofs. I did not find a result that proves convergence for deep policy gradient with fictitious play in zero-sum Markov games.

- Deep Reinforcement Learning from Self-Play in Imperfect-Information Games (Heinrich and Silver, 2016). Neural fictitious self-play (NFSP) learns approximate Nash equilibria of poker games end to end by combining fictitious self-play with deep reinforcement learning; ordinary reinforcement learning diverged on the small game (Leduc poker) where NFSP moved toward equilibrium.
- Policy Optimization in Zero-Sum Markov Games: Fictitious Self-Play Provably Attains Nash Equilibria. A smooth fictitious self-play whose best-response step is a PPO step with entropy regularization reaches a neighbourhood of a Nash equilibrium at a sublinear rate in zero-sum Markov games (a guarantee for the regularized method, not the exact equilibrium).
- Policy Optimization for Markov Games: Unified Framework and Faster Convergence (Zhang et al., 2022). Each iteration updates the policy at every state with a matrix-game algorithm and then the values; the state-wise average policy converges to an approximate Nash equilibrium of two-player zero-sum Markov games.
- Fast Policy Extragradient Methods for Competitive Games with Entropy Regularization (Cen, Wei and Chi, 2024). Entropy regularization gives linear-rate convergence to the regularized equilibrium

(including in zero-sum Markov games), and tuning the weight approaches an approximate Nash equilibrium of the unregularized game.

- Fictitious Play in Markov Games with Single Controller (Sayin, Zhang and Ozdaglar, 2022). A variant of fictitious play converges in identical-interest Markov games where one player controls the transitions; together with earlier results it covers both zero-sum and identical-interest Markov games.
- Fictitious play in zero-sum stochastic games (Sayin, Parise and Ozdaglar, 2022). A fictitious-play variant that combines classical fictitious play with Q-learning for two-player zero-sum stochastic games.